Class 07:Machine learning 1

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##background

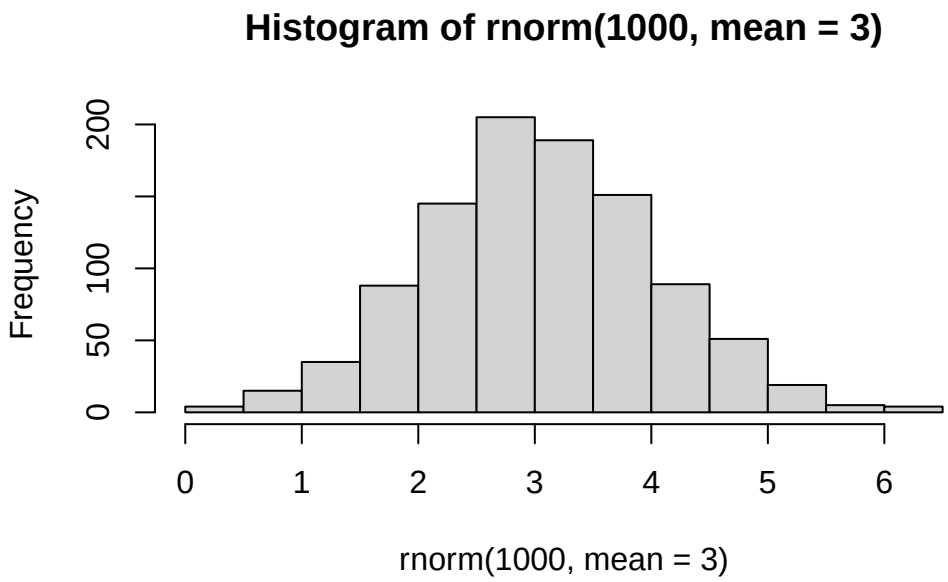
Today we will explore some core machine learning methods that are very popular in bioinformatics. These include **clustering** and **dimensionality reduction**

k-means clustering

The main function in the "base" R for K-means clustering is called `kmeans()`

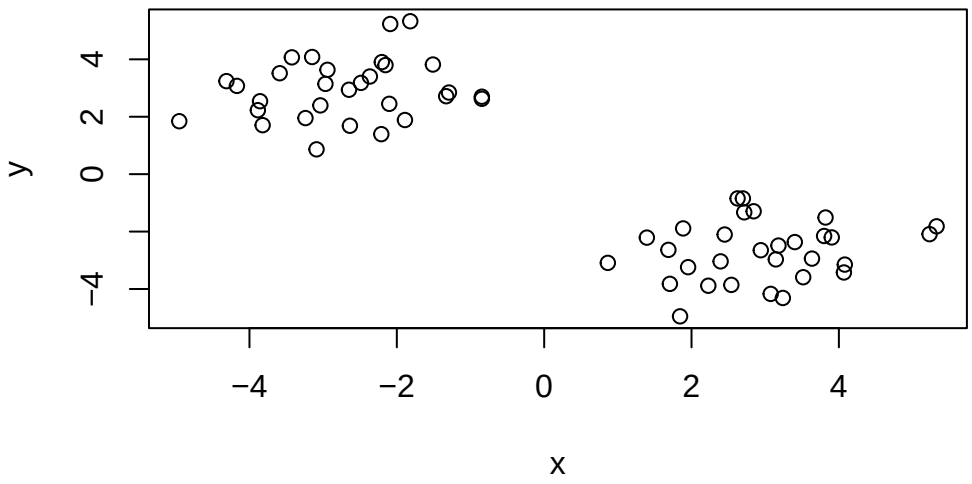
Before we go too deep let's make up some "simple" data that we can cluster and know if we are getting a good answer or not. To do this we can use `rnorm()` function:

```
hist ( rnorm(1000, mean = 3))
```




```
x <-c( rnorm(30, mean= -3), rnorm(30, mean= +3) )
```

```
z <- cbind(x=x,y=rev(x))
plot(z)
```




```
#cbind() / rbind() combines by colum/rows
```

```
p <- 1:5

cbind(p,rev(p))
```

```
     p
[1,] 1 5
[2,] 2 4
[3,] 3 3
[4,] 4 2
[5,] 5 1
```

```
rbind(p,rev(p))
```

```
  [,1] [,2] [,3] [,4] [,5]
p    1    2    3    4    5
     5    4    3    2    1
```

Now we can run `kmeans()` on this input `z` and see what the results look like

```
km <- kmeans(z,centers = 2)
km
```

```
K-means clustering with 2 clusters of sizes 30, 30

Cluster means:
          x         y
1  2.939620 -2.695773
2 -2.695773  2.939620

Clustering vector:
 [1] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 1 1 1 1 1 1 1 1
[39] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1

Within cluster sum of squares by cluster:
[1] 63.00271 63.00271
 (between_SS / total_SS =  88.3 %)

Available components:

[1] "cluster"      "centers"      "totss"        "withinss"     "tot.withinss"
[6] "betweenss"    "size"         "iter"         "ifault"
```

```
attributes(km)
```

```
$names
[1] "cluster"      "centers"      "totss"        "withinss"     "tot.withinss"
[6] "betweenss"    "size"         "iter"         "ifault"

$class
[1] "kmeans"
```

Q. How many points are in each cluster?

```
km$size
```

```
[1] 30 30
```

Q. What 'component' of your result object details cluster assignment/membership?

```
km$cluster
```

```
 [1] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 1 1 1 1 1 1 1 1
[39] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
```

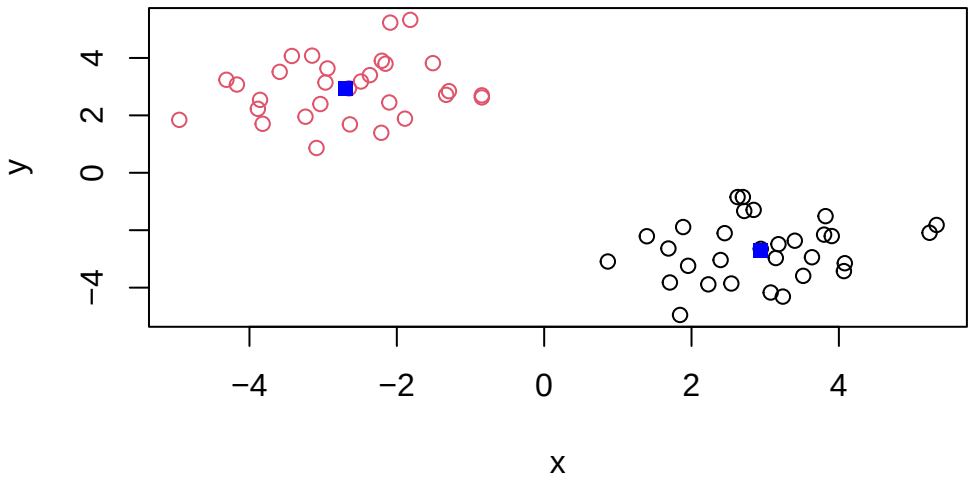
Q What 'component' of your result object details cluster center?

```
km$centers
```

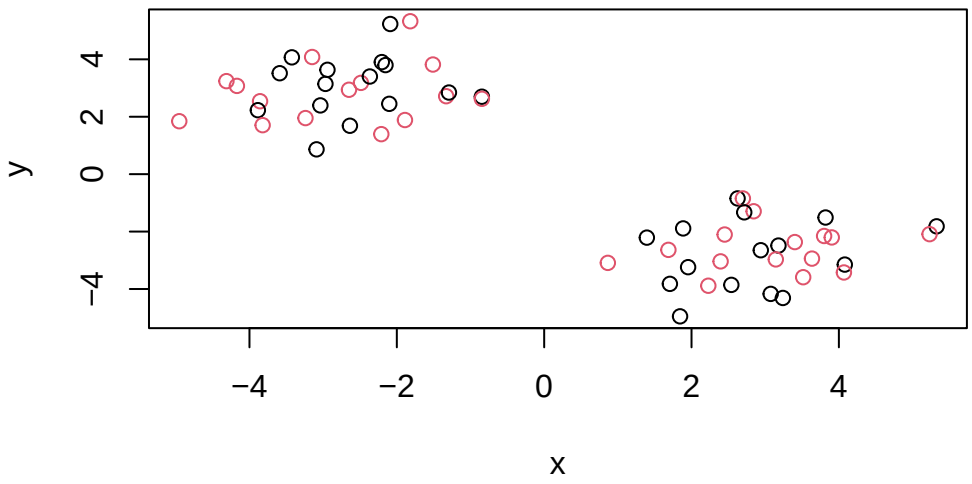
```
          x         y
1  2.939620 -2.695773
2 -2.695773  2.939620
```

Q. Plot `z` colored by the kmeans cluster assignment and add cluster centers as blue points

```
plot (z, col=km$cluster)
points (km$centers, col="blue", pch= 15)
```

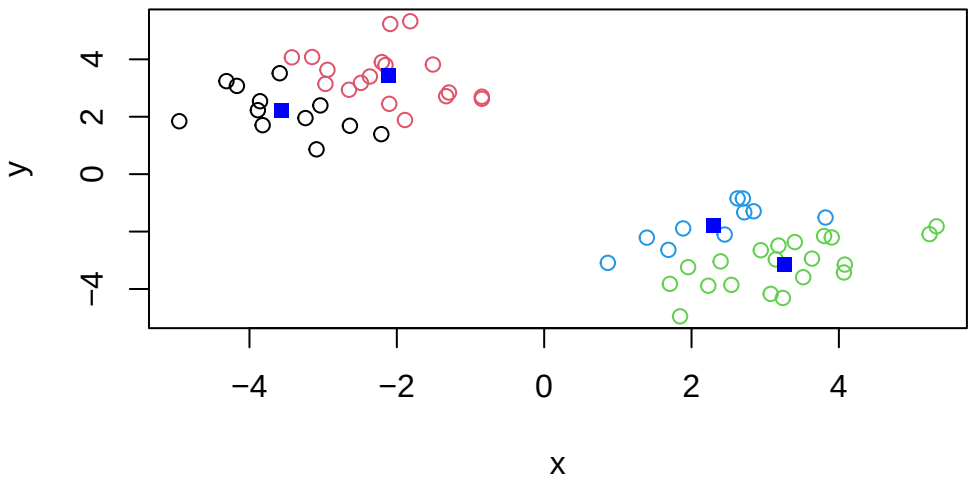



```
plot(z, col= c (1,2))
```



Q. Run a K-means clustering and plot the results asking for 4 clusters (k=4)?

```
km4 <- kmeans(z, centers =4)
plot(z, col=km4$cluster)
points(km4$centers, col="blue", pch=15)
```

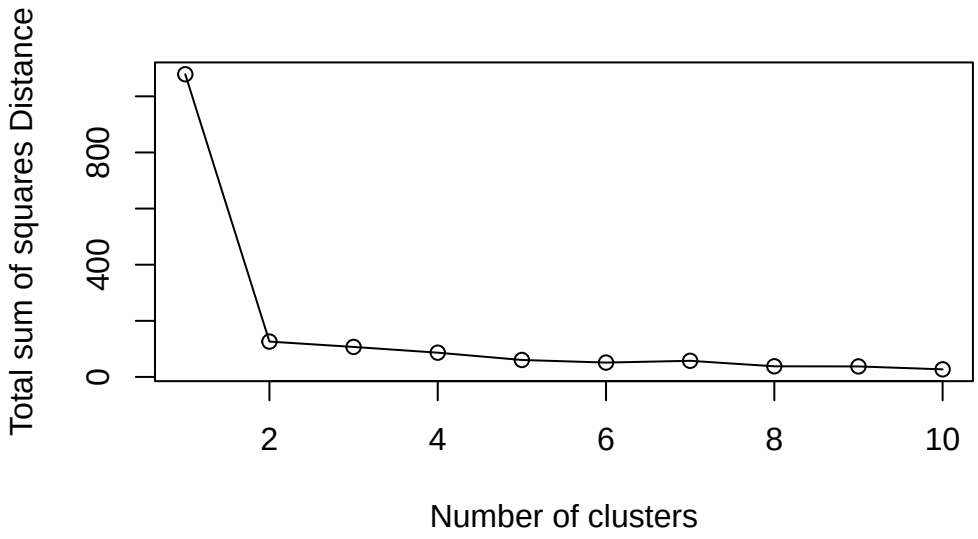


N.B. You need to tell k-means the number of clusters (i.e set `centers=2`)!!

One approach is to try different values for `centers` and then pick the best…

```
ans<-NULL
for(i in 1:10) {
  km <- kmeans(z, center = i)
  ans <- c(ans, km$tot.withinss)
}

plot (ans, type = "o",
      xlab = "Number of clusters",
      ylab = "Total sum of squares Distance" )
```



Hierarchial Clustering

The main function in "base" R for Hierarchical clustering is called `hclust()`

This function does not take your "raw" data for clustering, you must first build a "distance matrix" from your data and pass this as input to `hclust()`

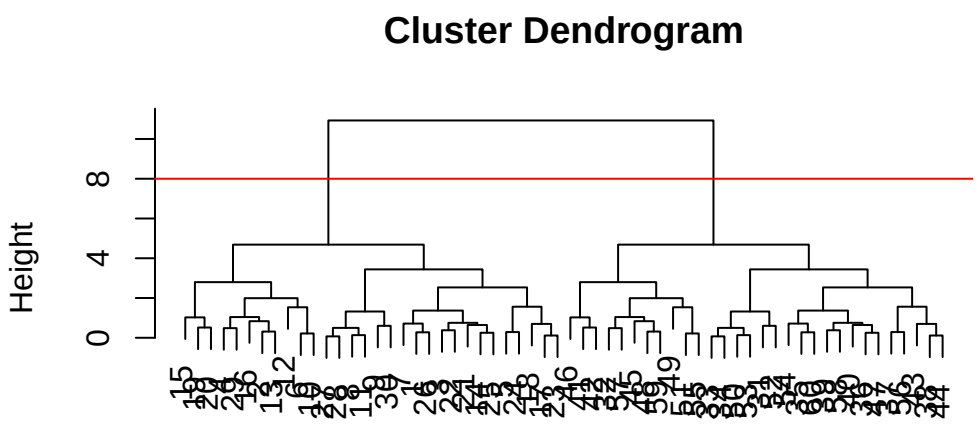
```
d <- dist(z)
hc <- hclust(d)
hc
```

```
Call:
hclust(d = d)

Cluster method   : complete
Distance         : euclidean
Number of objects: 60
```

There is a bespoke `plot()` method for `hclust` results objects.

```
plot(hc)
abline(h=8, col="red")
```



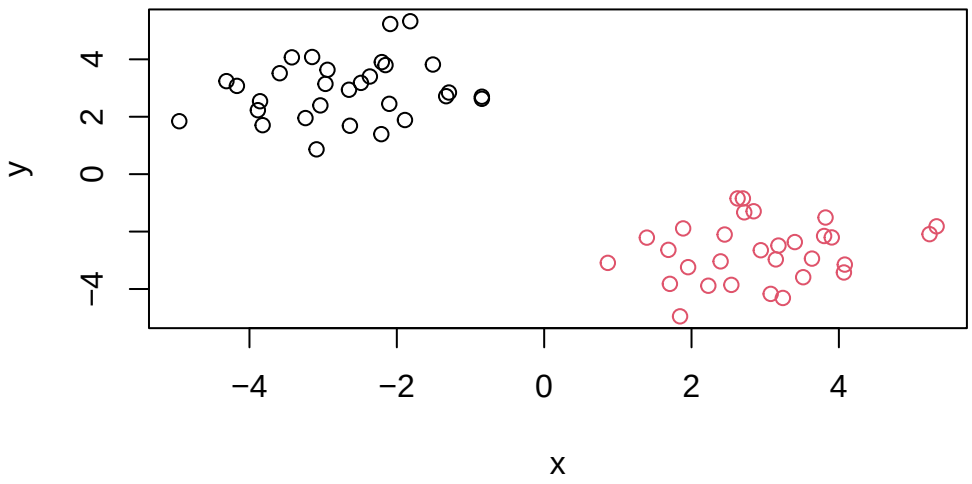
Once we have our `hclust` object (our "tree" of "cluster dendrogram") we can *"cut"* the tree to reval the clustering pattern.

```
cutree(hc, h=8)
```

```
 [1] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 2 2 2 2 2 2 2 2
[39] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
```

Q. Make a plot of `z` with your hclust results (i.e.l colored by cluster membership)

```
grps <- cutree(hc, k=2)
plot(z, col = grps)
```



##Principal Component Analysis (PCA)

PCA is a dimensionality reduction method that is popular for revealing patterns in complex datasets

Analysis of UK food data

let's look at some data on the eating habbits of folks from the UK to see if there are patterns and trends that have some regions being distinct from others.

##Data Import

The data is made available in CSV format so we can use three `reas.csv()` function for the import to R:

```
url <- "https://tinyurl.com/UK-foods"
x <- read.csv(url)
x
```

```
                   X England Wales Scotland N.Ireland
1             Cheese     105   103      103        66
2       Carcass_meat     245   227      242       267
3         Other_meat     685   803      750       586
4               Fish     147   160      122        93
```

```
5       Fats_and_oils      193   235      184       209
6               Sugars     156   175      147       139
7      Fresh_potatoes      720   874      566      1033
8           Fresh_Veg      253   265      171       143
9           Other_Veg      488   570      418       355
10 Processed_potatoes      198   203      220       187
11      Processed_Veg      360   365      337       334
12        Fresh_fruit     1102  1137      957       674
13            Cereals     1472  1582     1462      1494
14          Beverages       57    73       53        47
15        Soft_drinks     1374  1256     1572      1506
16   Alcoholic_drinks      375   475      458       135
17      Confectionery       54    64       62        41
```

Q1.How many rows and columns are in your new data frame named x? What R functions could you use to answer this questions?

```
nrow(x)
```

```
[1] 17
```

```
ncol(x)
```

```
[1] 5
```

#Preview the first 6 rows

```
x <- read.csv(url, row.names=1)
head(x)
```

```
              England Wales Scotland N.Ireland
Cheese            105   103      103        66
Carcass_meat      245   227      242       267
Other_meat        685   803      750       586
Fish              147   160      122        93
Fats_and_oils     193   235      184       209
Sugars            156   175      147       139
```

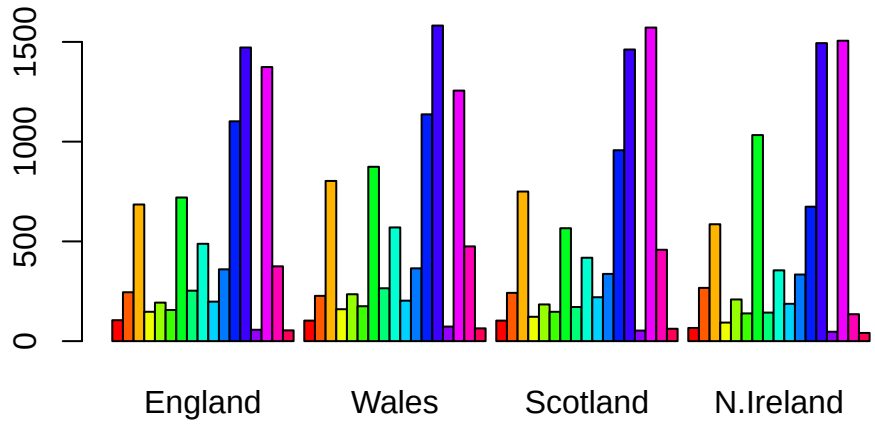
Q2. Which approach to solving the 'row-names problem' mentioned above do you prefer and why? Is one approach more robust than another under certain circumstances?

I prefer read.csv(url, row.names= 1) because it's cleaner and fixes the row names during import. It's better when you already know the first column should be row names. The manual method is useful if the files was already read incorrectly and needs to be fixed afterward.

Q3. Changing what optional argument in the above barplot() function results in the following plot?

The optional argument changed is beside, from TRUE to FALSE (or removed). setting beside = FALSE produces a stacked barplot instead of side-by-side bars.

```
# using base R
barplot(as.matrix(x), beside = T, col = rainbow(nrow(x)))
```



Q4. Changing what optional argument in the above ggplot() code results in a stacked barplot figure?

```
library(tidyr)

#convert data to long format for ggplot with pivot_longer()
x_long <- x |>
          tibble::rownames_to_column("Food") |>
          pivot_longer(cols = -Food,
                       names_to = "Country",
```

```
                        values_to = "Consumption")

dim(x_long)
```

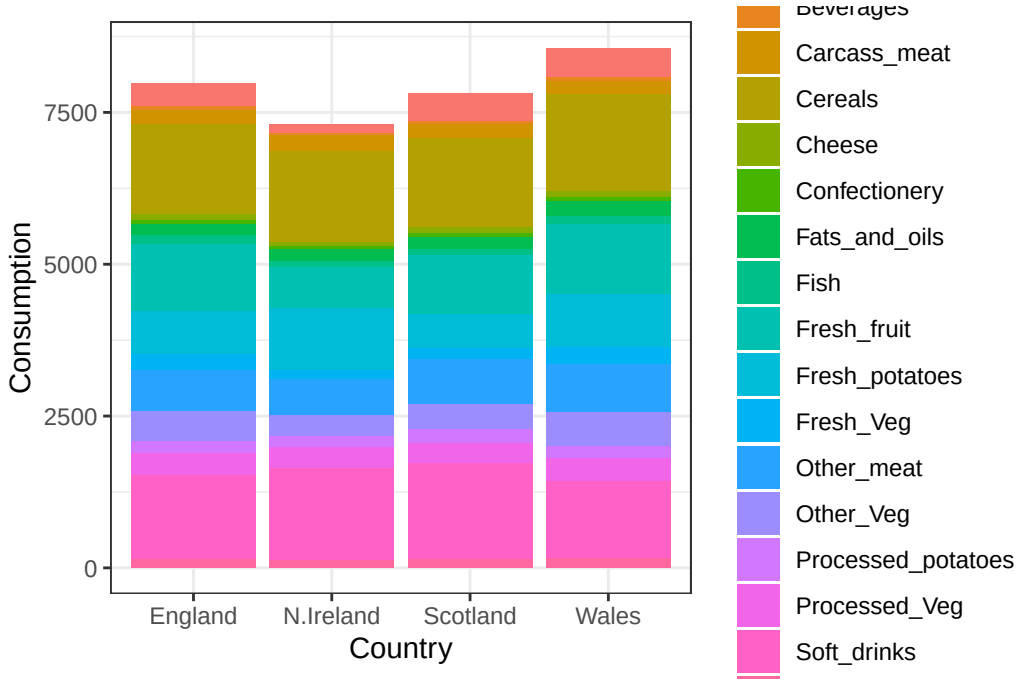
```
[1] 68  3
```

```
head(x_long)
```

```
# A tibble: 6 x 3
  Food            Country   Consumption
  <chr>           <chr>           <int>
1 "Cheese"        England           105
2 "Cheese"        Wales             103
3 "Cheese"        Scotland          103
4 "Cheese"        N.Ireland          66
5 "Carcass_meat " England           245
6 "Carcass_meat " Wales             227
```

```
library(ggplot2)
```

```
ggplot(x_long)+
  aes(x= Country, y= Consumption, fill= Food) +
  geom_col(position= "stack") +
  theme_bw()
```



The optional argument changed is position from "dodge" to "stack", which creates a stacked barplot instead of grouped bars.

##Tidy Data

Fix anything that went wrong with the data import.

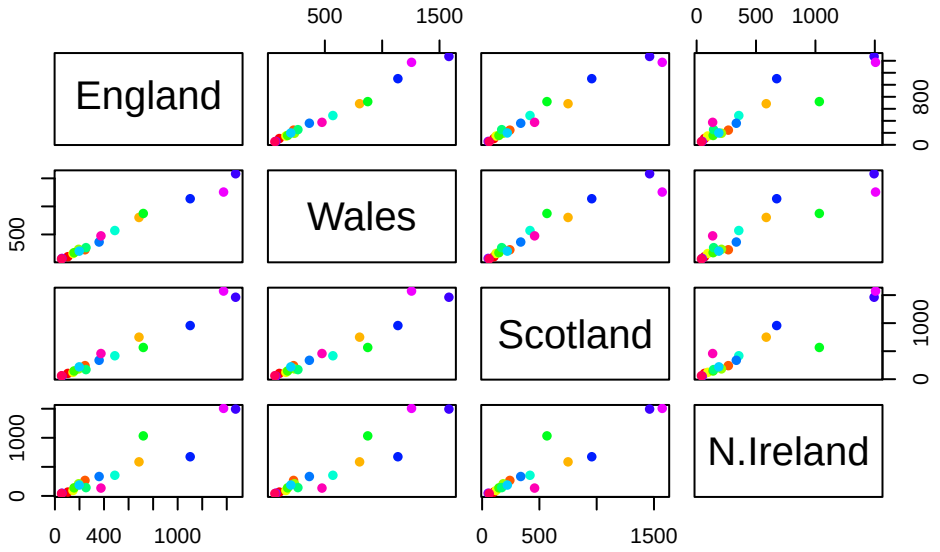
Exploratory analysis

Make some plots to help make sense of obvious trends...

> Q5: We can use the pairs() function to generate all pairwise plots for our countries. Can you make sense of the following code and resulting figure? What does it mean if a given point lies on the diagonal for a given plot?

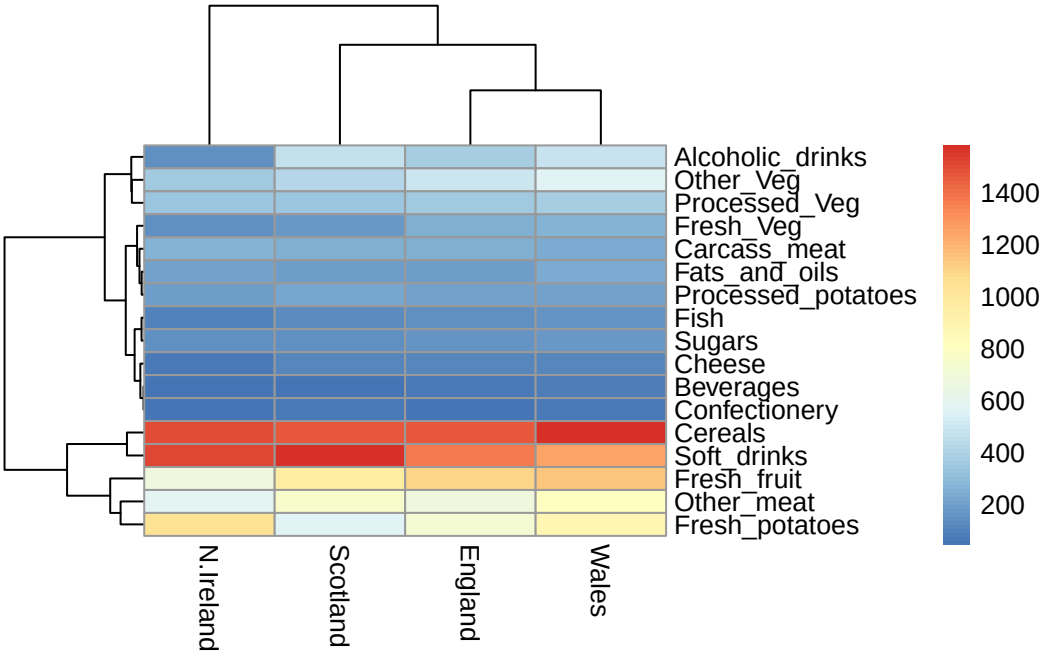
The pairs() function creates scatterplots comparing each pair of countries, where each point represents a food category. If a point lies on the diagonal, it means the consumption values for that food are equal (or very similar) between the two countries.

```
pairs(x, col=rainbow(nrow(x)), pch=16)
```




```
library(pheatmap)

pheatmap(as.matrix(x))
```



key point : Even relatively small datasets can prove challenging to interpret

Q6. based on the pairs and heatmap figures, which countries cluster together and what does this suggest about their food consumption patterns? Can you easily tell what the main differences between N. Ireland and the other countries of the Uk in terms of this data-set?

England, Wales, and Scotland cluster together, showing similar consumption patterns, N. Ireland is more distinct. While we can see it differs overall, the exact food differences aren't immediately clear from these plots alone

##PCA to the rescure

The main function in "base R for the PCA is called `prcomp()`. This function wants the the"observations" to be rows and the "variables" to be columns

So here we need to take the transpose of our `x` input object

```
pca <- prcomp( t(x) )
summary(pca)
```

```
Importance of components:
                            PC1      PC2      PC3       PC4
Standard deviation     324.1502 212.7478 73.87622 2.921e-14
Proportion of Variance   0.6744   0.2905  0.03503 0.000e+00
Cumulative Proportion    0.6744   0.9650  1.00000 1.000e+00
```

The returned `pca` object has components that can use to make our main result figures:

```
attributes(pca)
```

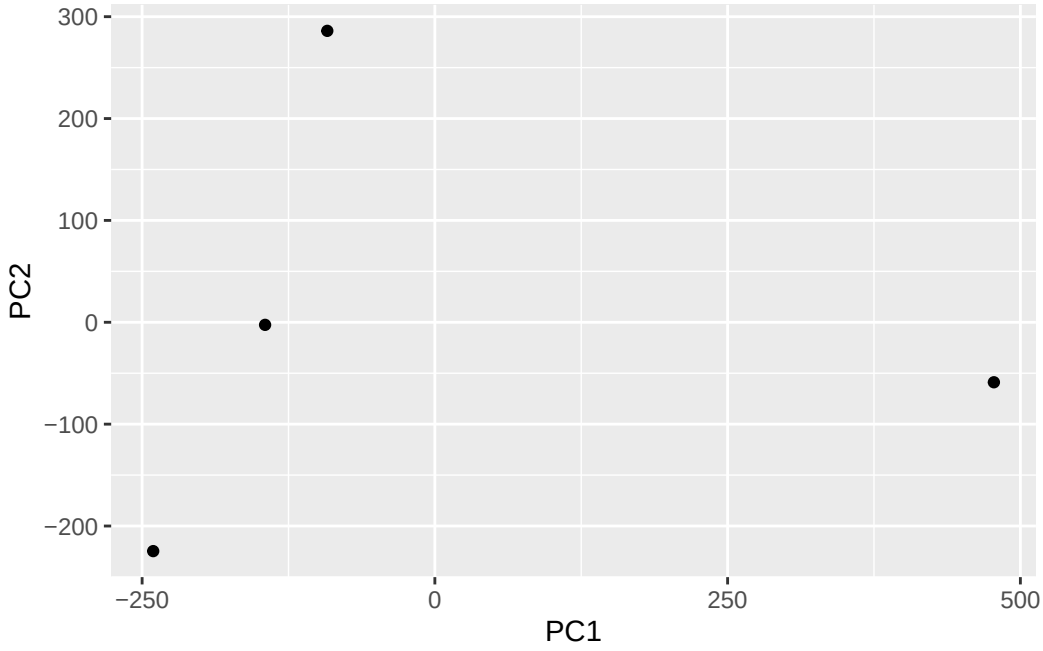
```
$names
[1] "sdev"     "rotation" "center"   "scale"    "x"

$class
[1] "prcomp"
```

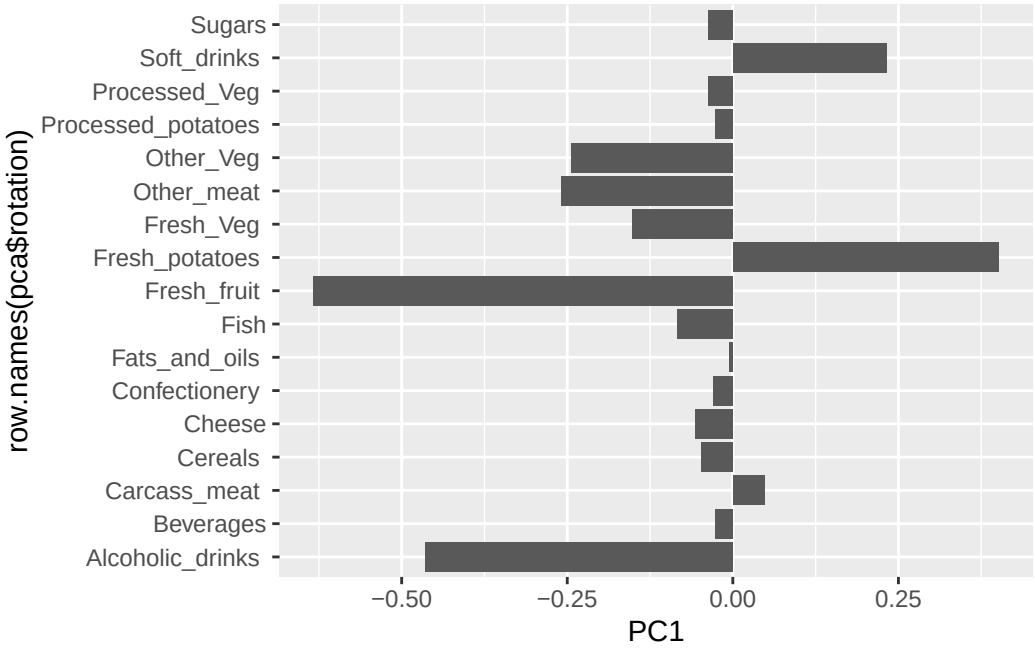
The main result figure from this analysis is called a "PC score plot" or "ordination plot" or "PC plot" or "PC1 vs PC2 plot". This plot shos us how samples (in thsi case countries) relate to each other along our new PC axis. This is our new "reduced-dimensional space". In this case 2 dimensions, PC1 and PC2, that capture most of the variance in the original 17 dimensional data-set.

```
library(ggplot2)

ggplot(pca$x) +
  aes(PC1, PC2) +
  geom_point()
```

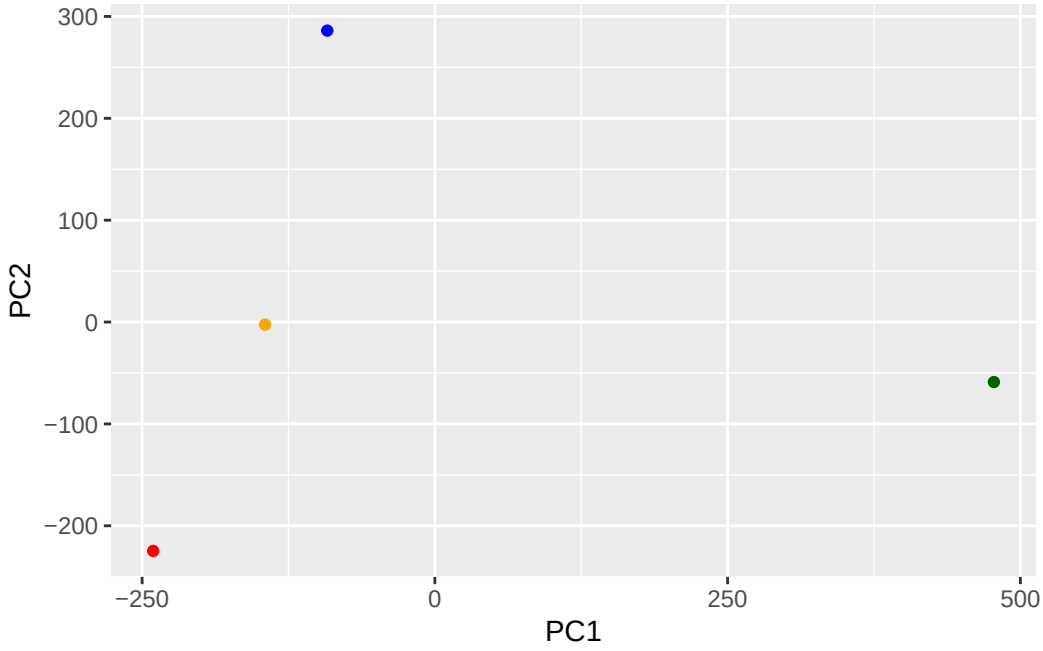



```
ggplot(pca$rotation) +
  aes(PC1, row.names(pca$rotation)) +
  geom_col()
```

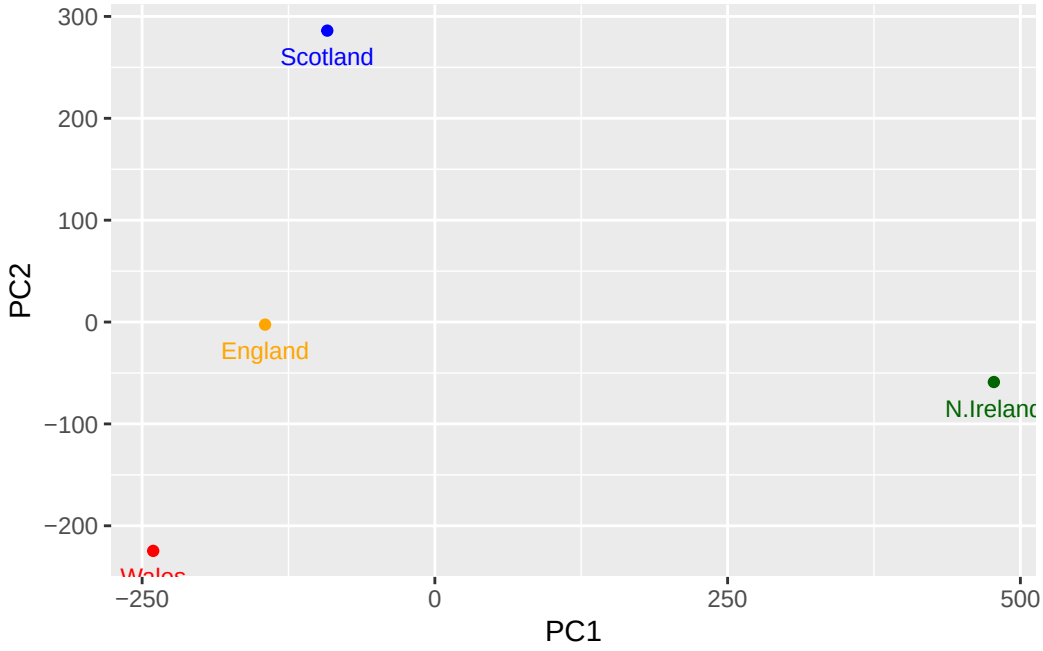



```
mycols <- c("orange", "red", "blue", "darkgreen")

ggplot(pca$x) +
  aes(PC1, PC2) +
  geom_point(col=mycols)
```



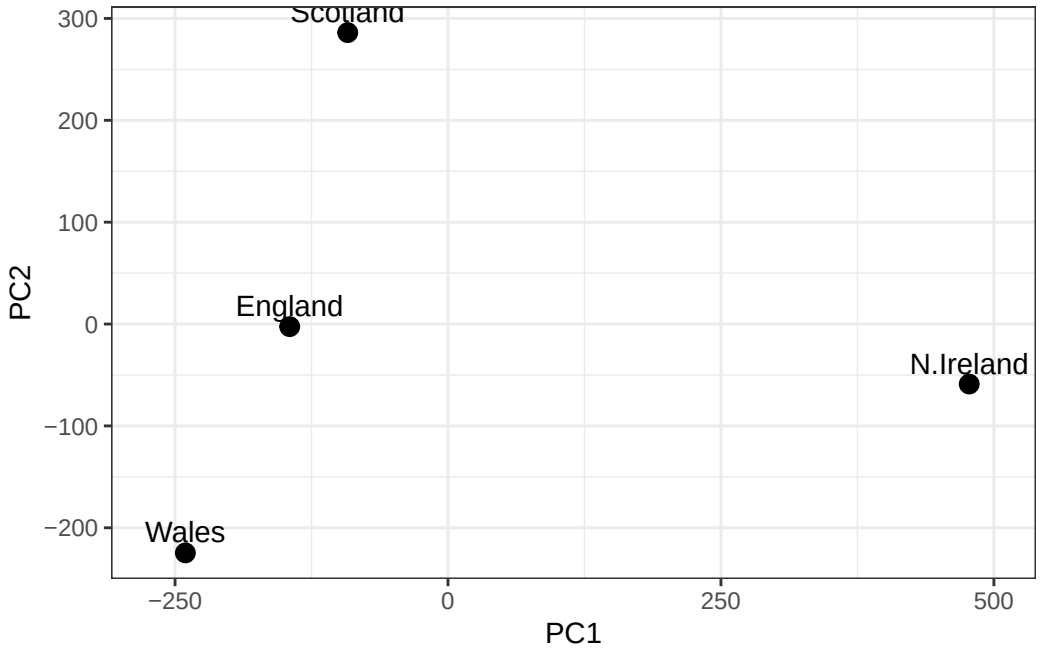

```
ggplot(pca$x) +
  aes(PC1, PC2, label=row.names(pca$x)) +
  geom_point(col=mycols) +
  geom_text(size=3, vjust=2, col=mycols)
```



Q7. Complete the code below to generate a plot of PC1 vs PC2. The second line adds text labels over the data points.

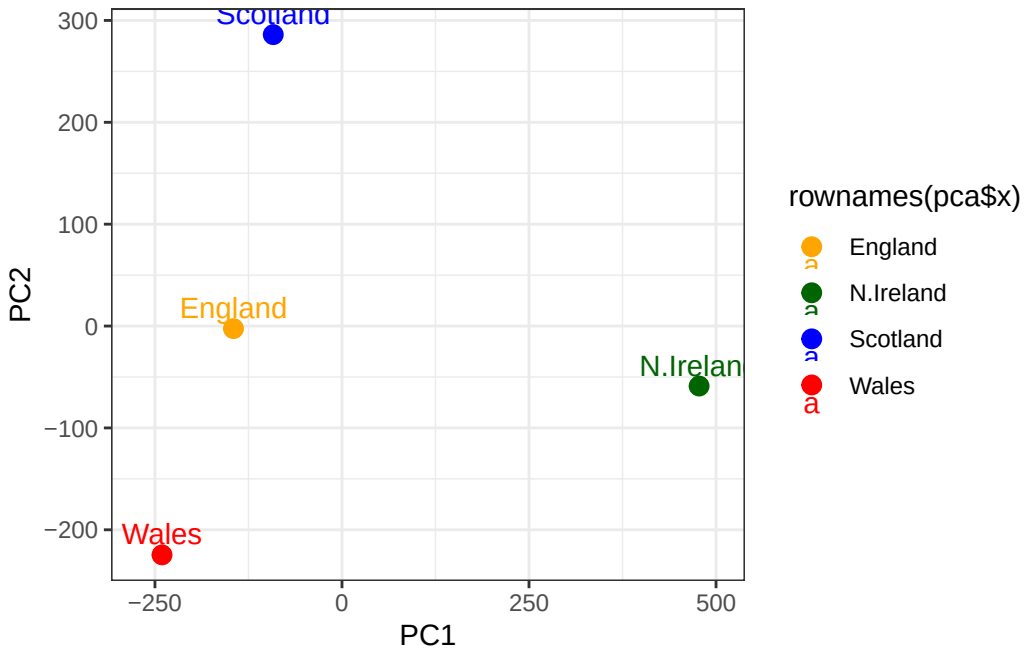
```
# create a data frame for plotting
df <- as.data.frame(pca$x)
df$Country <- rownames(df)

#plot PC1 vs PC2 with ggplot
ggplot(pca$x) +
  aes(x= PC1, y=PC2, label=rownames(pca$x)) +
  geom_point(size=3) +
  geom_text(vjust= -0.5) +
  xlim(-270, 500) +
  xlab("PC1") +
  ylab("PC2") +
  theme_bw()
```



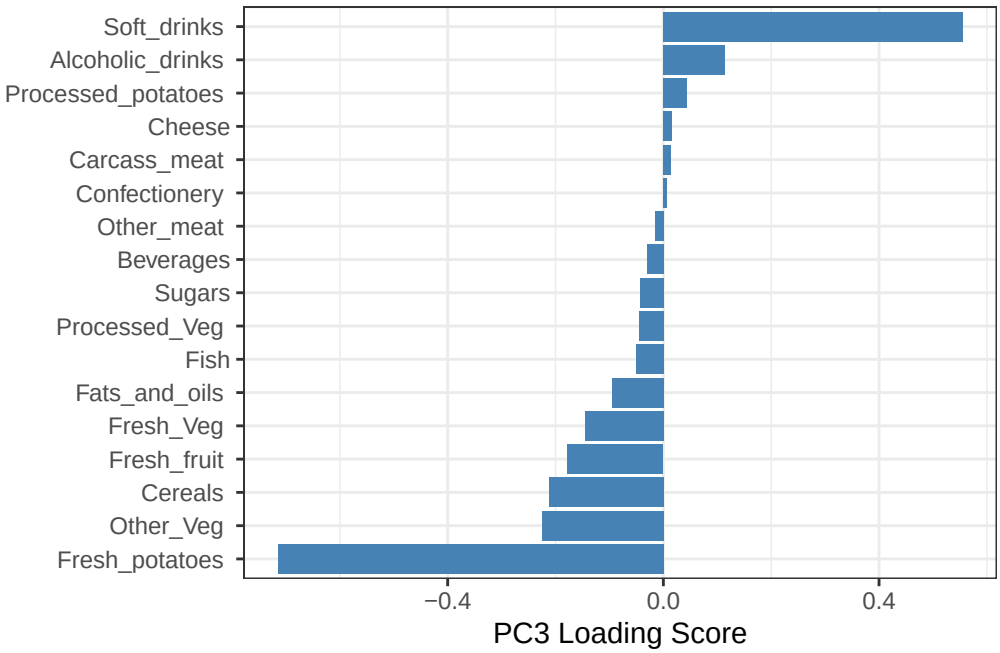
Q8. Customize your plot so that the colors of the country names match the colors in our UK and Ireland map and table at the start of this document.

```
ggplot(pca$x) +
  aes(x= PC1, y=PC2, label= rownames(pca$x), color = rownames(pca$x)) +
  geom_point(size= 3) +
  geom_text(vjust= -0.5) +
  scale_color_manual(values=c(
    "England" = "orange",
    "Wales"= "red",
    "Scotland" = "blue",
    "N.Ireland"= "darkgreen"
  )) +
  xlim(-270, 500) +
  xlab("PC1") +
  ylab("PC2") +
  theme_bw()
```



Q9. Generate a similar 'loading plot' for PC2, what two food group feature prominately and what does PC2 mainly tell us about?

```
ggplot(pca$rotation) +
  aes(x= PC2, y= reorder(rownames(pca$rotation), PC2)) +
  geom_col(fill= "steelblue") +
  xlab("PC3 Loading Score") +
  ylab("") +
  theme_bw() +
  theme(axis.text.y= element_text(size=9))
```



Fresh_veg and fresh_fruit 9strong loading one one side) vs soft drinks (an sometimes processed foods) on the opposite side. PC2 mainly represents a contrast between healthier fresh foods (fruits and vegetables) and less healthy / processed items (like soft drinks)

The score plot shows how samples are positioned along principal components, while the loading plot shows which variables influence those components. The loadings explain why samples appear where they do in the score plot.

Q10. How many genes and samples are in this data set? how many PCs do you think it will take to have a useful overview of this data set?

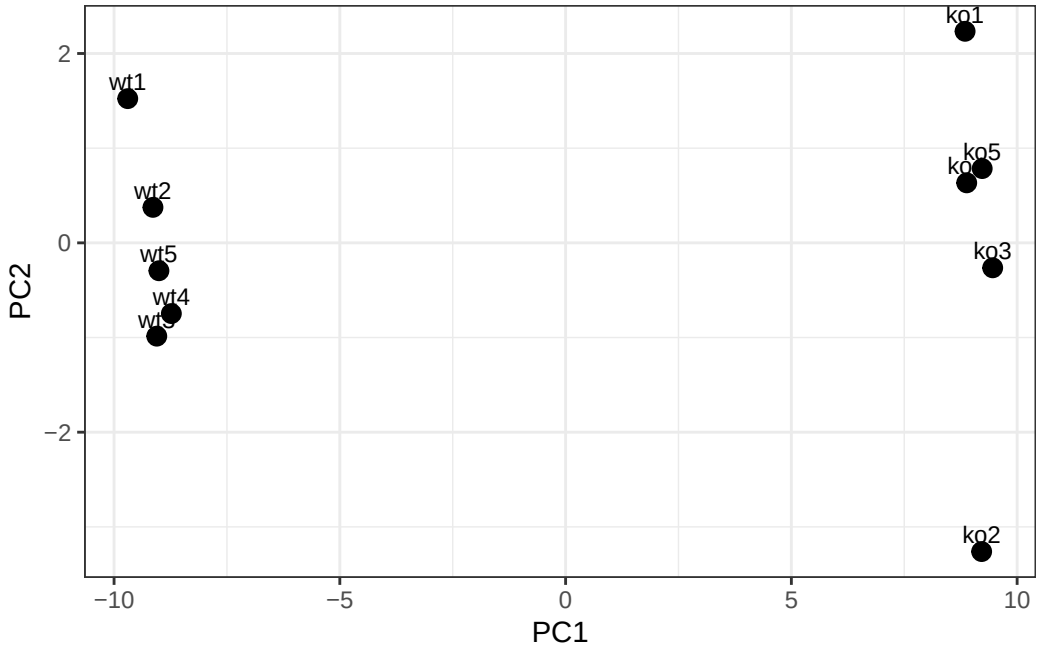
```
url2 <- "https://tinyurl.com/expression-CSV"
rna.dna <- read.csv(url2, row.names=1)
head(rna.dna)
```

```
       wt1 wt2  wt3  wt4 wt5 ko1 ko2 ko3 ko4 ko5
gene1  439 458  408  429 420  90  88  86  90  93
gene2  219 200  204  210 187 427 423 434 433 426
gene3 1006 989 1030 1017 973 252 237 238 226 210
gene4  783 792  829  856 760 849 856 835 885 894
gene5  181 249  204  244 225 277 305 272 270 279
gene6  460 502  491  491 493 612 594 577 618 638
```

```r
## Again we have to take the transpose of our data
pca <- prcomp(t(rna.dna), scale=TRUE)

#Create data frame for plotting
df <- as.data.frame(pca$x)
df$Sample <- rownames(df)

##Plot with ggplot
ggplot(df) +
  aes(x = PC1, y = PC2, label = Sample) +
  geom_point(size=3) +
  geom_text(vjust= -0.5, size=3) +
  xlab("PC1") +
  ylab("PC2") +
  theme_bw()
```



The data set has 100 genes and 10 samples. about 1-2 principal components are sufficient, since PC1 explains 92.6% of the variance and PC2 adds a small amount more.